Which Factors Contribute to the Accuracy and Efficiency of Horse Racing Betting Markets?

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ABSTRACT

This is the abstract.

- A brief introduction to the objectives.
- A brief description of the main work.
- A brief description of the contributions, major findings, results achieved and conclusion.

Acronyms

BSP — Betfair Starting Price
CDA — Continuous Double Auction
CLOB — Central Limit Order Book
DPM — Dynamic Pari-Mutuel Market
HKJC — Hong Kong Jockey Club
IP — In-play (after the off)
PNL — Profit and Loss
PP — Pre-play (before the off)
WAP — Weighted Average Price

Glossary

Arbitrage — A risk-free opportunity to make money
Back — A bet that something will happen
Daily Double — A bet on the winner of two consecutive races
Each-way — A combination of a win and place bet
Exacta — A bet on two horses which must come first and second

Lay — A bet that something will not happen

Options Contract — A right but not an obligation to buy or sell a security at a given date.

Pandas — Python library for data manipulation

Parlay — A single bet which links multiple consecutive wagers

Place — Come first or second in a race

Prediction Market — A market that facilitates bets on the outcomes of future events

Python — Programming Language

Show — Come first, second or third in a race

Tipster — Professional who publishes betting guides

1. Introduction

Horse racing markets have been academically studied as valuable research targets since at least the 1940s (Griffith, 1949). Racing markets are good to study for a few reasons. For one, they involve a large number of bettors. The bettors have varying levels of expertise ranging from absolutely zero to professional punters. These markets also closely mirror more ‘‘traditional’’ financial markets, and so research involving racing can be extrapolated out to the wider context of finance. Betting contracts are similar to options contracts in that they have a set expiry time and a payout determined by an external factor. The facets of horse racing make it an ideal ground to examine the efficient markets model (Snyder, 1978).

As of late, prediction markets have become a hot topic. Services such as Polymarket and Kalshi are offering the opportunity to bet on many world events. There has been research to suggest that prediction markets are more accurate than other forecasting techniques including polls (Berg, 2008). Horse racing markets are also fundamentally prediction markets, and have a long history of widespread use. As a prediction market, horse races provide good data since there are so many and the approximate date and time is known before hand. It is easier to determine their accuracy in this context as there are many repeated similar markets, unlike predictions for political events which can be very far apart.

Existing research into

1.1. Motivation

I was motivated to research this area due to the recent popularity of prediction markets as a tool for forecasting. As well as this, the regulation surrounding prediction markets is currently heavy within the UK. This is because they fall under the jurisdiction of the Gambling Commission. Many of the newly popular prediction market sites are currently illegal to use within the UK, but established exchanges such as Betfair are completely legal. Another interesting factor of being classed as gambling means that winnings within the UK are not taxed. This means that an equivalent strategy within a gambling market and the stock market may prove ultimately more profitable in the gambling market as the only fees to be paid are commission to the exchange. In the case of Betfair, commission can be even less than just 2% of successful bets.

1.2. Scope

This project observes a variety of different classifications of horse races. Each classification type is examined to see if there is any link between the classification of a horse race and its related betting market's accuracy and efficiency. In order to achieve this, a database of horse races is compiled using data from the Betfair exchange.

1.3. Aims

The primary aim of this project is to determine which factors are most significant in influencing the accuracy and efficiency of horse racing markets. There are many different ways to classify horse races including but not limited to time, distance and race type (Hurdles, Chase, Handicap, etc.). Additionally, races can be classified by their market factors such as trade volume or overround.

Accuracy of the market is a measure of how well the listed odds represent the true odds of a horse winning. In a market that was 100% accurate, the odds given would provide a 0% return on all winning bets. To determine the accuracy of the odds, I will be comparing various test statistics such as the Brier score and Log-Loss.

An efficient market is one in which prices are said to ''fully reflect'' all of the available information and are shown to be *weak-form* efficient when this can be demonstrated using an information set made exclusively of historical price data (Fama, 1970). A systematic betting strategy using historical pricing data that is able to generate a consistent profit demonstrates a lack of *weak-form* efficiency within a market (Hausch, 1981). In order to determine which groups of races are *weak-form* efficient, I shall apply various betting strategy profiles to them in isolation. Some of these strategies will simple and some will be adapted from previous work.

1.4. Objectives

- Perform a systematic review of existing research to determine existing standards for market accuracy and efficiency. Identify whether racing markets are generally considered to be accurate or efficient.
- Obtain an appropriate dataset for research.
- Categorise the data into potentially useful groupings.
- Examine each grouping to determine whether or not they fit the previously identified definitions of accurate or efficient.

2. Background

2.1. Bookmaking

2.1.1. Fixed-Price Odds

A fixed-price bookmaker will quote prices (odds) at which they are willing to sell bets. If the bet pays out, the bettor receives the return originally quoted to them. The bettor is given a guarantee of their winnings before placing the bet. It is important to note that it is unlikely a bookmaker will allow a bettor to ''lay'' a bet, that is to bet that something *will not* happen. Bookmakers make their money by appropriately setting odds such that in all cases (or at least over a period of time), they will ensure a profit. This is done by changing odds to account for the

amount of money bet already (balancing the books) and by providing unfair odds that do not sum to one.

2.1.2. Parimutuel Odds

Parimutuel betting (commonly known as a ‘tote’ in the UK) is a system in which money for all bets is placed into a pool and then distributed to winning bettors. This means that bettors are not given an exact payout quote at the time of purchasing their bet, but they may be given an estimate. Tote operators make money by taking a certain percentage of the pool for themselves, typically called the ‘‘track take’’. These types of bets are typically facilitated by the track operator themselves, such as the HKJC (Hong Kong Jockey Club, 2026).

2.1.3. Betfair Exchange

Betfair is a UK based company which operates a betting exchange service. A betting exchange is different from a typical bookmaker in that instead of the operator quoting odds prices or offering a tote service, it allows users to form bets with one another. Betfair does this using a CLOB, allowing users to either back (long) a bet or lay (short) one. Within the context of a horse race, this means that the win market (where bets are paid out only if the horse places first) features a different asset for every selection (runner) in the race. They can be thought of as largely analogous to options contracts, where the layer agrees to pay the backer a certain sum of money dependent on the outcome of the race.

A unique feature of the Betfair racing market is its "starting price" betting. Bettors who use the starting price option have their bets matched into the market at the off (when the race begins). They are also able to set minimum odds to back or maximum odds to lay. The price determination maximises volume and gives backers and layers the same price. In order to do that, existing positions within the CLOB may be matched against as well. Starting price bets aren’t guaranteed to be matched if the odds limit is at or above the starting price (Betfair,).

2.2. Literature Review

2.2.1. Accuracy of Horse Racing Markets

2.2.1.1. Measurement of Forecast Accuracy

One of the earliest methods for scoring the accuracy of a forecast is provided by Brier (1950). The Brier score is determined as a result of many predictions using the mean squared difference between the estimated probability and the actual outcome. Formally:

$$BS = \frac{1}{N}\sum_{t=1}^{N}(f_t - o_t)^2$$

where f_t is the forecast probability, o_t is the actual outcome (0 if did not happen or 1 if did) and N is the number of forecasts. In this system, a perfect forecast is given a score of zero. This can only be achieved if the forecast guesses with 100% certainty every time. Ranked Probability Score (Epstein, 1969) is an extension of the Brier score to support multiple outcomes Ranked Probability Score with binary outcomes can be shown to be equivalent to the Brier score (Weigel, 2007). Brier score will be the most useful in this report, as in (most) betting markets there are only two possible outcomes: a winning bet or a losing bet.

Good (1952) developed a method for scoring estimates of probabilities now known as log-loss. This method has since become popular as a scoring mechanism for machine learning models as it highly punishes overconfidence in incorrect predictions. For a multiple class classification problem:

$$LL = -\sum_{t=1}^{N} o_t \log(f_t)$$

where f_t is the forecast probability, o_t is the actual outcome (0 if did not happen or 1 if did) and N is the number of forecasts. A horse race only has one winner and so for a single race win market, it can be shown that $LL = -\log(f_{winner})$. Log-loss has become incredibly popular as a cost function (function that scores prediction accuracy) for machine learning. Machine learning textbooks (Goodfellow, 2016; Murphy, 2022) share that log-loss should be considered the standard cost function. Since log-loss easily supports binary outcomes, it can be applied in addition to Brier score as a measurement of accuracy for betting predictions. Because every non winning horse contributes the same score of zero, it appears to me that Brier score is more appropriate for general ranking of horse race win forecasting as it incorporates more information from the losing horses.

Wunderlich and Memmert (2020) demonstrated that the betting returns of a strategy are not necessarily indicative of accuracy in a forecasting model, despite demonstrating profitability. Instead, there is a difference in a model forecasting accurately and profitably.

2.2.1.2. Use of Markets for Forecasting

Current research shows prediction markets to provide accurate estimates of outcomes for elections over both long and short time frames (Berg, 2008). Rothschild (2009) showed that, when used appropriately, prediction markets can perform significantly better than the very best polls. Another study found prediction markets to outperform a survey when used to predict the reproducibility of studies published in psychology journals (Dreber, 2015).

Because of the demonstrated favourite-longshot bias in horse racing, many studies attempt to convert odds given by bookmakers or markets into ‘‘real’’ probabilities. Štrumbelj (2014) used Shin’s (1993) model to create more accurate forecasts than those created through odds normalisation or regression. These methods are criticised by Clarke et al. (2017) for potentially producing negative probabilities and in the case of normalisation, not accounting for the favourite-longshot bias. They share that an alternative power method instead performs as well as Shin’s, or better.

Strijbis and Arnesen (2019) find that the setup of the market influences accuracy the most, while features of the event and the market’s participants have an insignificant impact. Prediction markets require the right incentives in order to produce the most accurate and up to date forecasts (Chen, 2010). Markets will also suffer if a group of insiders are able to influence the outcome. If the goal of the market is to produce the best forecast, then it must incentivise immediate truth-telling instead of deception. Pennock (2004) suggests an alternative ‘‘dynamic pari-mutuel market’’ (DPM) which functions similarly to typical parimutuel betting. A DPM allows investors to buy and sell bets at a changing rate determined by the market, giving them an opportunity to hedge or sell their bets early.

2.2.1.3. Favourite-Longshot Bias

The favourite longshot-bias has been demonstrated within racing markets as far back as 1949 (Griffith, 1949). This is a phenomenon in which bettors tend to prefer to bet on outlier horses, as opposed to the safer options. In turn, this moves the odds given by the market as favourite horses are given worse implied probabilities of winning and longshots are given better implied probabilities of winning. Other than the long and short-odded horses, Griffith found that the odds for horses were on average accurate.

Some studies have attempted to determine why this is the case. Weitzman (1965) used utility theory to derive a utility curve for one bettor and found it to be similar to the theoretical one proposed by Markowitz. To follow this study, Ali (1977) observed over 20,000 races and found the favourite-longshot bias to be present. Ali creates a wealth utility function and determines that the average bettor values risk, thereby causing the discrepancy in prices. Subsequent research has also conjectured that bettors have a preference for risk at the racetrack. Kanto et al. (1992) found this to be the case using a data sample from Finland.

Golec and Tamarkin (1998) suggested an alternative explanation for the favourite-longshot bias. They claimed that it was possible that bettors could actually be risk-averse and instead have a preference for skewness. Skewness here is the asymmetry of betting returns about the mean. To back this up they examined popular betting strategies and determined that they were better aligned with a preference for skewness rather than risk. Another theory is put forward by Snowberg and Wolfers (2010) which states that the bias can be accounted for by the misperceptions of risk by the bettors.

The bias has also been shown to not be universal. In contrast to the evidence found in North American markets, Busche and Hall (1988) found that bettors at the Hong Kong Jockey Club do not exhibit the same apparent risk-loving behaviour. They posit that the larger betting volume in Hong Kong may influence the bias.

2.2.2. Efficiency of Horse Racing Markets

2.2.2.1. Measurement of Market Efficiency

Fama (1970) set out three tests of efficiency for markets depending on how rigourously they can be described as efficient. An efficient market is one that is said to fully incorporate all the available information into the prices. This means that, for an efficient market, it should be impossible to create a profit generating strategy. For a market to be *weak form* efficient, the information set is constrained to that of historical price data. The *semi-strong form* tests also incorporate information widely available to the general public. In the *strong form* tests, the insight of investors with insider knowledge is also considered.

A significant amount of research into the study of efficiency in racing markets is specifically focussed on parimutuel markets. This is the most common market for racing in the United States as well as Hong Kong. Comparitively little exists for markets using a CDA such as Betfair.

2.2.2.2. Relationship of Exotic Bets

A lack of market efficiency can be demonstrated by comparing the prices of exotic betting options with equivalently produced parlays. Exotic bets involve betting on a combination of different outcomes for different horses within one wager. Popular examples include the daily

double, in which the first two winning horses of the day must be predicted correctly, and an exacta, where the first two horses of a race must be predicted in order. A parlay is similarly a wager of multiple outcomes, however can be constructed from multiple seperate bets with the winnings from earlier bets accumulated into future ones.

Certain exotic bets can be shown to have an equivalent payoff function to an appropriately constructed parlay. For the market to be considered efficient, both the exotic bet and the parlay must have an equivalent price. This difference in a typical financial market may be considered an arbitrage opportunity as one would be able to buy at the low price and sell at the high price, guaranteeing a profit. Within parimutuel markets however this is not possible, as bets cannot be bought and sold at different prices.

Ali's (1979) analysis of 1,089 racing days uses this principle to compare the prices of daily doubles and their equivalently constructed parlays. In this analysis, Ali finds that the efficient market hypothesis could possibly hold for racing markets as the difference in price between the exotic bets and the parlays is not statistically significant. In a different study however, Asch and Quandt (1987) find that for both exacta and daily double bets, their payoffs are not the same as equivalent bets on individual horses, thus demonstrating market inefficiency.

It is uncommon for betting exchanges to offer any betting options for horse races outside of win, place and each-way. This means that a comparison of exotic bets and parlays is not possible. While it may be possible to compare the prices of the win and place bets to the each-way bets, the limited volume of non-win markets makes this difficult.

2.2.2.3. Betting Strategies

The most obvious way to demonstrate a lack of market efficiency is to produce a simple, profitable betting strategy.

In a study of 1,730 races in American parimutuel win markets, Snyder (1978) showed that because of the favourite-longshot bias, prices were indeed more favourable for the favourite horses, but the track take was too significant for a strategy of favourite betting to be profitable. This indicates some level of *weak form* efficiency, however without the track take present (and betting habits remaining the same), it would be certainly impossible to describe the market as informationally efficient. Snyder also demonstrated that this bias exists among betting experts (or ''tipsters'') which makes it seem also unlikely that the market is *strong form* efficient.

Hausch et al. (1981) set about creating a technical betting system for place and show markets. Their strategy only used historical pricing data and so attempts to demonstrate a lack of *weak form* efficiency. They argued that their system successfully exploited a market inefficiency and was able to generate a consistent profit. With a follow up paper, Hausch and Ziemba (1985) greatly improved the system making it more profitable and applicable to any racetrack. They also demonstrated how more people using the system would ultimately cause the market to become efficient.

3. Objectives, Specification & Design

4. Methodology & Implementation

4.1. Data Acquisition

4.1.1. Betfair Exchange UK Horse Racing Win Markets

Historical odds pricing data is typically very difficult to find if not impossible from most bookmakers. This was a significant factor in choosing to use prices from the Betfair exchange. Betfair provides historical betting data at multiple levels of resolution including a generous free tier. The data provided in this set is structured like exchange data, with multiple entries per race documenting changes in prices over time. For my purposes, this is beyond what I need as I am only interest in the BSP and not the price evolution. I used this dataset at first but found this is beyond what I need for my purposes, as I am only interested in the BSP and not the price evolution.

Betfair also provides a directory of starting price files and allows users to filter them by location and market type. This dataset contains some useful information not included in the free tier of the standard historical data files such as trade volume and minimum and maximum prices for each selection. It is also significantly larger, containing data from as far back as 2008, compared to 2016 with the other set.

I produced a Python script to automatically download all of the UK win market files from this second dataset and process them to be used within Pandas. The script saves the data to binary files for quick retrieval. It makes sure to not redownload files it has already obtained so it can easily update the existing database. As well as this, it automatically slows down when rate limited to ensure that all files are downloaded.

The final dataset contains over 170,000 races from 24,000 different events in the UK, starting from September 1st 2008. There are over 1.5 million entries for selections within the races. This data is formatted into three different tables: events, races and, runners.

4.1.1.1. Events

Field	Data Type	Notes
`event_id`	`Integer`	Unique identifier provided by Betfair
`location`	`String`	
`event_date`	`Datetime`	

4.1.1.2. Races

Field	Data Type	Notes
`race_id`	`Integer`	Unique identifier provided by Betfair
`race_type`	`String`	Indicates the format of the race: handicap, hurdles, chase, etc.
`race_time`	`Datetime`	
`distance_f`	`Integer`	Furlongs are one eigth of a mile long.
`race_no`	`Integer`	Races are given numbers indicating their sequence position within an event.
`event_id`	`Integer`	Relational key.

4.1.1.3. Runners

Field	Data Type	Notes
selection_id	Integer	This identifier is unique per horse, not per entry.
selection_name	String	
win_lose	Boolean	Win=True, Lose=False
bsp	Float64	Betfair Starting Price
implied_odds	Float64	1/BSP
ppwap	Float32	Pre-play weighted average price.
morningwap	Float32	Weighted average price of bets before 11AM GMT
ppmax	Float32	Pre-play max price.
ppmin	Float32	Pre-play min price.
ipmax	Float32	In-play max price.
ipmin	Float32	In-play min price.
morningtradedvol	Float32	Trade volume before 11AM
pptradedvol	Float32	Trade volume before the off
iptradedvol	Float32	Trade volume after the off
race_id	Integer	Relational key.

4.1.2. Hong Kong Jockey Club

I also downloaded data from the website of the Hong Kong Jockey club in order to compare it against the UK market. This was done by scraping their webpages for historical racing events. Ultimately it proved that this data was not particularly useful, however. The available data I was able to collect only went back to 2022 and the reduced number of racecourses operated by the HKJC meant that the number of races was only around 2000, compared to the 170,000+ of the UK dataset.

4.2. Software Development

I decided to use Python through Jupyter Lab to write and run my software as it is what I had the most experience with for statistical analysis.

4.2.1. Data Structure and Statistics

4.2.1.1. Pre-Processing

I pre-processed my data to include some more fields that would be useful. Pre-processing fields like this means that the calculations do not need to be repeated later on.

The first field to be added was a normalised probability field. This is useful as the implied odds given by the starting price (almost) never sum to be one. The sum of implied odds given is known as the ‘‘overround’’. An overround greater than one for a back market shows that there is negative arbitrage and a weighted selection of back bets is guaranteed to lose money. Since the data for this project is using the BSP, it can trivially be shown that it is possible to arbitrage almost every single market. This is because for BSP, both back bettors and lay bettors are given the same price. In real terms this arbitrage is actually impossible to achieve however, as the BSP is not possible to know before the off, and by betting this way you are certain to move the market.

I then added fields for partial pre calculation of the Brier score and Log-Loss. This is particularly useful as it means that to determine the scores for a certain subset of races, they can simply be summated using the groups generated in the next section.

4.2.1.2. Categorisation

Given the available data from my dataset I decided upon a list of ways to categorise the horse races: race number, distance, hour of day, pre-play trade volume, race type (handicap, hurdles, etc.), location, number of runners and, overround. To do this I produced a list containing these different classifications of races created using the `groupby` method in Pandas. The `groupby` method makes it easy to perform aggregate calculations per group created through the method. This means that a function such as `sum` will be ran per group, instead of for all data in the DataFrame.

4.2.2. Trading Strategies

Trading strategies are implemented using a system in which the strategy function receives a list of available runners to bet on and returns a ‘‘bet slip’’. The bet slip contains information about which race and runner it chose, as well as the amount bet and whether or not the selection won. The amount bet is able to be submitted as either an absolute value, or as a percentage of the currently held bank value.

The bet slips are passed to a function which actions the bets and determines the PNL for each one. It also sorts the bets over time to generate a timeline of events.

Each betting strategy is written using vector based functions. This allows the underlying library (Pandas or Numpy) to perform many operations in parallel. There are two primary benefits to this. Firstly, it significantly improves the readability of the code as the need for looping is removed. The code is much more representative of what is actually happening. Secondly, there is a major performance increase by writing the software in this way.

4.2.2.1. Naive Strategies

The first naive strategy simply bets one unit of currency on the favourite (runner with the highest implied probability) in every race. Due to how I set up the strategy implementation, this is incredibly simple as it just involves grouping the runners per race and then submitting a bet of one. If the favourite horse is given odds lower than its real odds, then this strategy should consistently make money. Breaking even is an implication of the favourite being accurately priced over time.

The other naive strategy is similar to the first, but instead of betting a constant amount on each favourite, it attempts to ‘‘hedge’’ the bet by betting lower for worse odds and higher for better odds. If the prices for the favourite horse are accurate, then this strategy should consistently break even.

4.2.2.2. Learning Strategy

I created a very simple learning strategy that uses a rolling average to estimate the current likelihood of the favourite horse winning. It assumes the current average of win rate of the favourite horse to be the true win rate and uses the Kelly Criterion to size a bet. That is: $f^* = (bp - q)/b$ where f^* is the fraction of wealth to bet, p is the likelihood of the bet winning, q is $1 - p$ and b is the proportion of wealth won if successful (Thorp, 2006). It is worth noting

that this does not account for the impact that a bet will have on the market. This is a very simple strategy which does not use any information outside of the historical prices. If this strategy is able to make a profit, it indicates that the market is not *weak-form* efficient.

5. Results, Analysis & Evaluation

5.1. Accuracy

5.2. Efficiency

6. Legal, Social, Ethical & Professional Issues

- Use of live animals?
- Profit Derived from gambling losers
- Similarity to stock market
- legality of companies betting

7. Conclusion

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